ASSIGNMENT

1. If $c_1, \ldots\ldots, c_n$ be positive real numbers, show that $\left(c_1+\ldots.+c_n\right)^3 \le n^2\left(c_1^3+\ldots.+c_n^3\right)$ When does the inequality reduce to equality ?

Sol If $a_1, \ldots.. a_n$, $b_1, \ldots.. b_n$, be real numbers, then by Cauchy Schwarz inequality,

$$\left(a_1b_1+\ldots+a_nb_n\right)^2 \le \left(a_1^2+\ldots+a_n^2\right)\left(b_1^2+\ldots.+b_n^2\right) \qquad \text{.... (1)}$$

Putting $a_i = C_i^{3/2}, b_i = c_i^{1/2}, \left(i=1,2,\ldots\ldots., n\right)$ in the above inequity, we have

$$\left(c_1^2+\ldots.+c_n^2\right)^2 \le \left(c_1^3+\ldots+c_n^3\right)\left(c_1+\ldots+c_n\right) \qquad \text{....(2)}$$

Again, putting $a_i = c_i$, $b_i = 1$, (i = 1,2,.....,n) in (1), we have

$$\left(c_1+\ldots+c_n\right)^2 \le n\left(c_1^2+\ldots+c_n^2\right) \qquad \text{...(3)}$$

Squaring both sides of (3) and using (2), we immediately have

$$\left(c_1+\ldots+c_n\right)^3 \le n^2\left(c_1^3+\ldots..c_n^3\right)$$

The above inequality reduces to an equality iff each of the inequalities (2) and (3) reduces to an equality ,i.e. iff

$$c_i^{3/2} : \ldots\ldots. : c_n^{.3/2} :: c_1^{1/2} : \ldots. : c_n^{1/2},$$

And $c_1 : \ldots.. : c_n = 1 : \ldots : 1$

i.e., iff $c_1 = c_2 \ldots. = c_n$

2. If a, b and c are the sides of a triangle and a + b + c = 2, then prove that $a^2 + b^2 + c^2 + c^2 + 2abc < 2$

Sol We know a + b + c = 2 and squaring, we get

$4 = (a + b + c)^2 = a^2 + b^2 + c^2 + 2\,(ab + bc + ca)$

$$\Rightarrow a^2+b^2+c^2 = 2\left(ab-bc-ca\right)$$

Adding 2abc to both sides, we get

$$a^2+b^2+c^2+2abc = 2\left(2-ab-bc-ca+abc\right)$$

To prove $a^2 + b^2 + c^2 + 2abc < 2$, it is enough to prove that

$$2\left(2-ab-bc-ca+abc\right) < 2$$

Or $\quad 2+abc-ab-bc-ca<1$

Or $\quad ab+bc+ca-abc-1>0$

$\because \quad$ a + b +c = 2s = 2

$\Rightarrow \quad$ s = 1

$$(1-a)(1-b)(1-c)=1-(a+b+c)+(ab+bc+ca)-abc$$

$$0\le(s-a)(s-b)(s-c)=1-2+(ab+bc+ca)-abc$$

$$\Rightarrow ab+bc+ca-abc-1>0$$

3. For n $\in$ N, n > 1, show that $\frac{1}{n}+\frac{1}{n+1}+\frac{1}{n+2}+...+\frac{1}{n^2}>1$

Sol We have

$$\underbrace{\frac{1}{n}+\frac{1}{n+1}+\frac{1}{n+2}+..+\frac{1}{n^2}}_{(n^2-n)\text{terms}}>\frac{\frac{1}{n}+\left(\frac{1}{n^2}+\frac{1}{n^2}+..+\frac{1}{n^2}\right)}{(n^2-n)\text{terms}}$$

$$\Rightarrow \frac{1}{n}+\frac{1}{n+1}+\frac{1}{n+2}+...+\frac{1}{n^2}>\frac{1}{n}+\frac{(n^2-n)}{n^2}=\frac{1}{n}+1-\frac{1}{n}=1$$

4. Show that $\left(1+\sqrt{\frac{1}{2}}+...\sqrt{\frac{1}{n}}\right)/\sqrt{n}\le(2n-1)^{1/4}$

Sol Applying Tcheby Chef's inequality to the sets of numbers $1,\frac{1}{2},....\frac{1}{n};1,\frac{1}{2},...\frac{1}{n},$ we obtain

$$\left(1+\frac{1}{2}+...+\frac{1}{n}\right)^2\le n\left(\frac{1}{1^2}+\frac{1}{2^2}+.....+\frac{1}{n^2}\right)$$

$$\le n\left[1+\frac{1}{1.2}+...+\frac{1}{(n-1)n}\right]$$

$$=n\left[1+\left(1-\frac{1}{2}\right)+....+\left(\frac{1}{n-1}-\frac{1}{n}\right)\right]=n\left(1+1-\frac{1}{n}\right)$$

Taking positive square roots of both sides, we have

$$\left(1+\frac{1}{2}+...+\frac{1}{n}\right)\le\sqrt{(2n-1)} \qquad ...(i)$$

Again, applying Tcheby Chef's Inequality to the sets of numbers

$1, \sqrt{\frac{1}{2}}......., \sqrt{\frac{1}{n}} : 1, \sqrt{\frac{1}{2}}, ..., \sqrt{\frac{1}{n}}$, we have

$$\left(1+\sqrt{\frac{1}{2}}+.....\sqrt{\frac{1}{n}}\right)^2 \le n\left(1+\frac{1}{2}+...+\frac{1}{n}\right) \qquad ...(ii)$$

From (i) and (ii), we have

$$\left\{1+\sqrt{\frac{1}{2}}+\sqrt{\left(\frac{1}{n}\right)}\right\}^2 \le n\sqrt{(2n-1)}$$

Therefore $\dfrac{\left(1+\sqrt{\frac{1}{2}}+....+\sqrt{\frac{1}{n}}\right)}{\sqrt{n}} \le (2n-1)^{1/4}$

5. If a, b, c are all positive and no two of them are equal, then prove that

(a) $a^3+b^3+c^3 > \frac{(a+b+c)^3}{9} > 3abc$ (b) $a^4+b^4+c^4 > abc(a+b+c)$

Sol **(a)** without any loss of generality we may assume that a < b < c. By applying the generalised Tcheby Chef's inequality to three sets numbers each of which is the same a, b, c we obtain

$$\frac{a^3+b^3+c^3}{3} > \frac{a+b+c}{3}\frac{a+b+c}{3}\frac{a+b+c}{3}$$

i.e. $a^3+b^3+c^3 > \frac{(a+b+c)^3}{9}$ ....(i)

Again, Since the arithmetic mean exceeds the geometric mean

$$\left(\frac{a+b+c}{3}\right)^3 > abc \qquad ...(ii)$$

i.e., $\frac{(a+b+c)^3}{9} > 3abc$ .....(a)

(b) As in (a), without any loss of generality we may assume that a < b < c. Since a < b < c, therefore, $a^3 < b^3 < c^3$. Applying Tcheby Che's inequity to the sets of number a, b, c, a^2, b^3, c^3, we obtain

$$\frac{a^4+b^4+b^4}{3} > \frac{a^3+b^3+c^3}{3}.\frac{a+b+c}{3} \qquad ...(iii)$$

Also, from (a) $\frac{a^3+b^3+c^3}{3} > abc$ ...(iv)

From (iii) and (iv) , we have

$$a^4+b^4+c^4>abc(a+b+c)$$

6. If a, b, c are positive and unequal, show that

$$\left(a^7+b^7+c^7\right)\left(a^2+b^2+c^2\right)>\left(a^5+b^5+c^5\right)\left(a^4+b^4+c^4\right)$$

Sol $\left(a^7+b^7+c^7\right)\left(a^2+b^2+c^2\right)-\left(a^5+b^5+c^5\right)\left(a^4+b^4+c^4\right)$

$$=\sum\left(a^7b^2+a^3b^7-a^5b^4-a^4b^5\right)$$

$$=\sum a^2b^2\left(a^5+b^5-a^3b^2-a^2b^3\right)$$

$$=\sum a^2b^2\left(a^3-b^3\right)\left(a^2-b^2\right)$$

The differences a^2-b^2, a^3-b^3 are both of the same sign, and therefore $\left(a^2-b^2\right)\left(a^3-b^3\right)$ is positive. Similarly, the other two terms in the above sum are also positive. Therefore

$$\left(a^7+b^7+c^7\right)\left(a^2+b^2+c^2\right)-\left(a^5+b^5+c^5\right)\left(a^4+b^4+c^4\right)>0$$

7. If a, b, c are positive and if p, q, r are rational numbers such that p – q – r (≠0) and r (≠0) have the same sign, then show that

$$\left(a^p+b^p+c^p\right)\left(a^q+b^q+c^q\right)\geq\left(a^{p-r}+b^{p-r}+c^{p-r}\right)\left(a^{q+r}+b^{q+r}+c^{q+r}\right)$$

Show that if either (i) a = b = c, or (ii) p = q + r, or (iii) r = 0, then equality holds

Sol $\left(a^p+b^p+c^p\right)\left(a^q+b^q+c^q\right)-\left(a^{p-r}+b^{p-r}+c^{p-r}\right)\left(a^{q+r}+b^{q+r}+c^{q+r}\right)$

$$=\sum\left(b^qa^p+a^qb^p-a^{p-r}b^{q+r}-a^{q+r}c^{p+r}\right)$$

$$=\sum a^qb^q\left(a^{p-q}+b^{p-q}-a^{p-q-r}b^r-a^rb^{p=q-r}\right)$$

$$=\sum a^qb^q\left(a^{p-q-r}-b^{p-q-r}\right)\left(a^r-b^r\right)$$

Since p – q – r and r have the same sigh, the differences $a^{p-q-r}-b^{p-q-r}$ and a^r-b^r have the same sign or are both zero Therefore,

$$a^qb^q\left(a^{p-q-r}-b^{p-q-r}\right)\left(a^r-b^r\right)\geq 0,$$

And similarly each of the other two terms in the above sum is also non-negative, so that the sum is non negative. This proves the inequality $a^qb^q\left(a^{p-q-r}-b^{p-q-r}\right)\left(a^r-b^r\right)\geq 0$ And similarly each of the other two terms in the above sum is also non-negative, so that the sum is non negative. This proves the inequality

Also, if any of the conditions is satisfied, then at least one of the factors in each term is $\sum a^q b^q \left(d^{p-q-r} - b^{p-q-r}\right)\left(a^r - b^r\right)$ vanishes and therefore the sum is zero. This proves that the equality holds

8. Let a, b, c be real numbers with 0 < a, b, c < 1 and a + b + c = 2. Prove that $\frac{a}{1-a}.\frac{b}{1-b}.\frac{c}{1-c} \geq 8$

Sol : Here we use A.M. – G.M.

$$a = \frac{(a+b+c)+(a+b+c)}{2} \geq \sqrt{(a+b+c)(a-b+c)}$$

$$b = \frac{(b+a-c)+(b-a+c)c}{2} \geq \sqrt{(b+a-c)(b-a+c)}$$

$$c = \frac{(c+a-b)+(c-a+b)}{2} \geq \sqrt{(c+a-b)(c-a+b)}$$

$$\therefore a.b.c = \frac{[(a+b+c)+(a-b+c)][(b+a-c)+(b-a+c)][(c+a-b)+(c-a+b)]}{8}$$

$$\geq \sqrt{(a+b-c)(a-b+c)(b+a-c)(b-a+c)(c+a-b)(c-a+b)}$$

$$= (2-2c)(2-2a)(2-2b)$$

$= 8(1-c)(1-a)(1-b)$ [∵ a + b + c = 2]

$$\therefore \frac{a}{1-a}.\frac{b}{1-b} - \frac{c}{1-c} \geq 8$$

9. If a, b, c are sides of triangle, show that $\left(1+\frac{b-c}{a}\right)^a \left(1-\frac{c-a}{b}\right)^b . \left(1+\frac{a-b}{c}\right)^c < 1$

Sol Since a, b, c are the sides of triangle

a + b – c > 0, b + c – a > 0 , c + a – b > 0 ...(1)

Thus, $1+\frac{b-c}{a}, 1+\frac{c-a}{b} - 1+\frac{a-b}{c}$ are all positive ...(2)

Take $1+\frac{b-c}{a}$.....'a' times

$1+\frac{c-a}{b}$.....'b' times

$1+\frac{a-b}{c}$.....'c' times

And apply A.M. – G.M. inequality

$$\frac{a\left(1+\frac{b-c}{a}\right)+b\left(1+\frac{c-a}{b}\right)+c\left(1+\frac{a-b}{c}\right)}{a+b+c}$$

$$>\left[\left\{1+\frac{b-c}{a}\right\}^{a}\times\left(1+\frac{c-a}{b}\right)^{b}\times\left(1+\frac{a-b}{c}\right)^{c}\right]^{\frac{1}{c+b+c}}$$

L.H.S of the inequality = 1

i.e. 1 > R.H.S

10. Prove that $\frac{1}{\log_2 \pi}+\frac{1}{\log_5 \pi}>2$

Sol Let $\log_2\pi$ **=** a and $\log_5 \pi$ = b

$\therefore \quad 2^{a}=\pi$ and $5^{b}=\pi$

$\therefore \quad 2=\pi^{1/a}$ and $5=\pi^{1/b}$

$\therefore \quad 2\times 5=\pi^{\frac{1}{a}+\frac{1}{b}}$

i.e., $\quad 10=\pi^{\frac{1}{a}+\frac{1}{b}}$

But 1 > π^2 (since $\pi=\frac{22}{7}$)

$\therefore \quad \pi^{2}<\pi^{\frac{1}{a}+\frac{1}{b}}$

$\therefore \quad 2<\frac{1}{a}+\frac{1}{b}$

or $\quad \frac{1}{a}+\frac{1}{b}>2$

Hence $\frac{1}{\log_2 \pi}+\frac{1}{\log_5 \pi}>2$